

Public-Artifact Sufficiency Audit of the SLRTP2025 Back-Translation Evaluator: A Failed Reconstruction and Descriptive Probe Analysis

Xin Ji^{1*} and Cheng Zhang¹

^{1*}Beijing Institute of Petrochemical Technology, Beijing, China.

Contributing authors: 2024540101@bipt.edu.cn;

Abstract

We audit whether the released SLRTP2025 back-translation (BT) evaluator for sign language production can be reconstructed from its public artifacts, an evaluation-only repository without training scripts. Our reconstruction attempt failed.

Twenty recipe-conditioned runs—fourteen legacy-implementation seeds and six near-faithful runs—reach dev BLEU 6.5–9.8 versus 13.38, teacher-forced token accuracy 55–60% versus 96.5%, and training-pool readout 7.2–9.3 versus 78.8. Because all constructible checkpoints are less competent, we cannot distinguish whether the non-reproduction of a constructed adversarial replay-probe response (+**10.24** sacreBLEU, post-selection descriptive interval) reflects the competence gap, an unrecovered training mechanism, or their interaction.

Reference replacement with human back-translations is associated with a gap reduction to +**0.95** (from +**10.03**) on a 461-item subset—a single-annotator point estimate that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects. Donor-pool substitution reverses the sign; matched-subset analysis (SMD<0.1) confirms a residual origin effect. These findings constitute a failed public-artifact reconstruction, not a causal checkpoint effect or general leaderboard instability. We frame the probe as an adversarial lexical-coupling stress test throughout.

Keywords: sign language production, back-translation evaluation, reproducibility audit, reference-frame sensitivity, learned evaluation metrics, PHOENIX-2014T

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-artifact sufficiency audit*: whether the released SLRTP2025 BT evaluator can be reconstructed from its public artifacts. We address three research questions:

- RQ1. Can the public artifacts reproduce the released evaluator’s competence and training-pool readout?
- RQ2. Does a frozen adversarial replay probe’s ordering reproduce across evaluator checkpoints constructed from the public recipe?
- RQ3. How sensitive is the probe response to the reference text and to donor origin?

We do *not* frame the work as a test of checkpoint identity. Every checkpoint we can construct from the public recipe is less competent than the released evaluator (14 paper-derived reconstructions: dev BLEU-4 6.5–9.4; 6 near-faithful runs: 8.3–9.8; broader constructible family training-log max 10.02; all vs. released 13.38), so competence and checkpoint identity are confounded. We therefore frame RQ2 strictly as a *descriptive non-reproduction*: among independently constructed but uniformly less-competent checkpoints, the ordering does not reproduce. Without a competence-matched independent implementation—which we could not obtain—we cannot distinguish whether this reflects the competence gap, an unrecovered training mechanism, or their interaction. The replay probe itself is an *adversarial lexical-coupling stress test*: the source-conditioned retriever selects donors by surface vocabulary overlap with the German source, which is identical to the scoring reference in PHX-public, so the probe deliberately targets the strongest form of the shortcut rather than a deployable retrieval pipeline.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026) (Hong and Košecká, arXiv preprint); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai et al. (2025); Jiang et al. (2025), yet BT remains central to SLRTP2025 Walsh et al. (2025). Chen et al. (2022) showed that learned BERT-based evaluation metrics can fail to reproduce due to undocumented preprocessing, missing code, and baseline implementation differences—a methodology directly relevant to our finding that our public-artifact reconstruction attempt failed to reproduce the released checkpoint. An unresolved question is whether a retrieval-reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by an adversarial retrieval-reference probe on PHX-public: under the released evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale) versus 12.78 for the recorded test poses. Because the

retriever selects donors by lexical overlap with the source sentence—which is identical to the scoring reference in PHX-public—this probe is a deliberately constructed adversarial stress test, not evidence of general evaluator preference or better signing.

We address the three research questions with the following evidence (full results in §4.2–§4.3.1). For RQ1, none of our reconstructions reached the released evaluator’s competence. For RQ2, all 36 non-degenerate constructible checkpoints give negative probe gaps; we cannot determine whether this reflects the competence gap or a checkpoint-identity effect. For RQ3, reference replacement with human back-translations is associated with a gap reduction; donor-pool substitution reverses the sign. The source-conditioned retriever shares surface vocabulary with the original reference (§3.2), so the RQ3 result conflates reference-text properties, lexical-coupling disruption, paraphrase shift, and BLEU floor effects.

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator’s training pool, so the adversarial probe isolates a risk that production systems face in diluted form. The audit contributes a transferable template (provenance table, checkpoint registry with SHA-256 deduplication, evaluator-replication controls, teacher-forced-vs-free-decode diagnostics) for public-artifact reproducibility testing. The specific probe response is a PHX-public released-evaluator case study and does not support general claims about BT evaluation or retrieval-based SLP.

2 Related Work

Learned evaluation in sign language production.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); BT was proposed as an SLP evaluation protocol by Saunders et al. (2021), who argued it shifts absolute scores but not relative rankings between similar configurations — a proposition our audit does not directly test (its similarity condition is unmet in our family) but re-examines for a released evaluator on which a retrieval-probe reversal appears. SignBLEU Kim et al. (2024), SILVER-Score Imai et al. (2025), and pose-native measures offer complementary views, but BT BLEU remains central to SLRTP2025 Walsh et al. (2025). Although newer metrics provide better human-correlated evaluation Jiang et al. (2025); Cory et al. (2026) (BackTranslation2.0, arXiv preprint), the SLRTP2025 BT evaluator is the *standardized infrastructure* on which the challenge ranking depends; auditing its reproducibility and failure modes is infrastructure maintenance, not a claim that decoded BT-BLEU is the best available metric.

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025’s leading entry is retrieval-based, making donor copying pertinent Walsh et al. (2025). Memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal

et al. (2022); Khandelwal et al. (2020); a shortcut under one learned evaluator does not show the ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) show BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low recorded-pose BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU; and model-based generation metrics are further criticized by Yazdani et al. (2026); Cory et al. (2026); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2026) released human back-translations and documented reference information loss, Alkain et al. Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora, and Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Reproducibility of learned evaluation metrics.

Chen et al. Chen et al. (2022) showed that reproducing BERT-based evaluation metrics can fail due to undocumented preprocessing, missing code, and baseline implementation differences, even when model weights are released—a finding methodologically close to our public-recipe non-reproduction result. The difference is that our audit targets a full BT evaluator pipeline (architecture, training recipe, and checkpoint) rather than a metric alone, and the non-reproduction is demonstrated across 36 non-degenerate independently trained instances.

PHOENIX-2014T as a language resource: known limitations.

PHOENIX-2014T Camgoz et al. (2018) (8,257 weather-forecast interpretations by 9 signers) is the most widely used DGS parallel corpus, but four properties constrain our audit’s generalizability: (i) template density inflates train–test n-gram overlap Alkain et al. (2026) (32 items whose reference exactly matches a training text account for a disproportionate share of the gap); (ii) reference texts are speech transcriptions, not DGS translations Czehmann et al. (2026); (iii) the 178-keypoint pose materialization discards handshape and non-manual detail (our MSKA control uses 133 keypoints; SI Sup. L); (iv) signer-dependent evaluation can overestimate capability Artiaga et al. (2025). The lexical coupling between source and reference amplifies these effects in our adversarial probe. The transferable audit components (provenance table, checkpoint registry, leakage controls) apply to any learned evaluator paired with a public training corpus.

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint, x_n the source sentence (German) for item n , and $o_{i,n}$ the pose sequence for output condition i and item n ; E_c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^N)$, where y_n is the reference text and

Table 1: Novelty relative to closely related PHOENIX-2014T and learned-metric reproducibility studies.

Study	Primary finding	Method	Distinction from present work
Czehmann et al. (2026)	Reference replace-ment changes BLEU scores and rankings on PHOENIX-2014T	Human back-translations of test set	We apply the matched-subset protocol to the released evaluator with a frozen replay probe and quantify paired attenuation
Alkain et al. (2026)	Highest train-test text overlap among tested SLT corpora	N-gram overlap quantification	We use their finding to contextualize why PHX-public is particularly susceptible to reference coupling
Artiaga et al. (2025)	Signer-dependent evaluation overestimates SLT capability	Signer-disjoint splits	We include signer as a regression covariate (not signer-disjoint evaluation; signer contributes $\leq 1\%$ of gap variance)
Jiang et al. (2025)	BT likelihood is a more stable correlate of pose/embedding/BT human judgment than decoded BLEU	Meta-evaluation of pose/embedding/BT metrics	We report that teacher-forced likelihood does not reproduce the reversal (REC NLL 3.060 vs PURE 3.479)
Chen et al. (2022)	BERT-based metrics fail to reproduce due to missing preprocessing/code	Multi-implementation replication	We extend the learned-metric reproducibility question to a full SLP evaluator pipeline with 70 training runs across 13 families
This work	Available public artifacts do not reproduce across 13 families, the released evaluator’s fixed-output cross-competence or replay-evaluator re-decoding, probe behavior	70 training runs across 13 families, fixed-output cross-evaluator re-decoding, reference/donor substitution	—

M is the corpus-level sacreBLEU functional unless stated otherwise. We call a result a *descriptive non-reproduction* when a probe response observed under the released evaluator does not appear across independently constructed checkpoints under a fixed output set; this term is descriptive and does not imply a causal checkpoint-identity effect (which we cannot test because all constructible checkpoints are less competent). The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval-generator compositions; oracle-gloss variants are diagnostic controls.

Operational definition of public-artifact sufficiency.

A publicly documented training recipe is *sufficient* when checkpoints constructed from it reach a predefined competence gate (our operational tolerance, not a community standard): $\text{dev BLEU-4} \geq 12.38$ and $\text{dev WER} \leq 86.37$ (one-sided). The gate conclusion is insensitive across a $[0.5, 2.0]_{\text{BLEU}} \times [1.5, 4.5]_{\text{WER}}$ grid (§4.3). The

Table 2: Three-way data provenance for PHX-public. The released evaluator’s training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs share the reason `absent_from_slrtp2025_released_pose_records` (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (.pt)	7,060	515	641
Released evaluator <code>config.yaml</code> manifest	7,060	515	641
Missing from official	36	4	1

source of the gap may reflect information incompleteness, stochastic underspecification, or implementation bias; without the original training command, these cannot be separated.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 2 gives the three-way provenance: the official PHOENIX-2014T split, the released pose materialization, and the released evaluator’s actual training manifest (confirmed via `config.yaml`, whose manifest matches the shipped .pt tensors). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L); the protocol summary is in SI Table S28.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution. The data flow is: official PHOENIX-2014T video at 25 fps → SLRTP2025 skeleton extraction at 25 fps → our `[:,2]` subsampling to 12.5 fps for evaluator input; the official evaluation command uses `--fps 25` (the input file’s native rate), but the evaluator was trained on 12.5 fps-subsampled poses, so our subsampling matches the training distribution rather than the command-line flag. A three-path regression confirms no double subsampling (details in SI Sup. F). PHX-public uses $\alpha=0.88$ (selected on dev); the supplementary splits use $\alpha=0.5$.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is 9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbafa301. Reproduction is supported at four layers (§ Sup. Q, Table S36): (i) **L1—recompute selected reported statistics**: a single command (`make core-audit`) regenerates the registry and verifies headline numbers from committed per-item decodes; it does *not* retrain models or decode from checkpoints, and does not verify every number in the paper; (ii) **L2—analysis re-execution**: each analysis script re-executes from the committed per-checkpoint data and hypotheses; (iii) **L3—model re-evaluation**: re-decoding fixed outputs from checkpoint weights requires the training bundle; (iv) **L4—from-scratch retraining**: full retraining requires the restricted PHOENIX-2014T licence, the exact environment, and approximately 200 GPU-hours. The `make core-audit` command executes L1 only and is not an end-to-end reproduction. A claim-to-file manifest mapping each headline number to its source file and verification command is in `artifact/claim_manifest.json`. An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N.

Lexical coupling of the source-conditioned probe (adversarial design).

In PHX-public, the source sentence x_n (German weather report) is identical to the reference text y_n^{orig} at the sequence level (the reference is a speech transcription of the same German sentence). The TN-PURE-v1 retriever selects training-pool donors by token-set Jaccard similarity between x_n and training texts, so it preferentially retrieves donors whose accompanying training texts share surface-level vocabulary with both the source *and* the reference. Scoring a donor copy against this lexically coupled reference under a learned evaluator therefore confounds donor similarity, evaluator readout, and reference overlap. We explicitly frame this probe as an *adversarial stress test* targeting the strongest form of the lexical-coupling shortcut—not as evidence of general evaluator preference, deployable retrieval quality, or better signing. This framing has two consequences: (i) the +10.24 gap is the expected output of a deliberately adversarial design, not a discovery about the evaluator’s ranking behavior; (ii) the human-reference attenuation analysis (§4.3.1) reflects disruption of the lexical coupling as well as reference-text information loss, paraphrase shift, and BLEU floor effects—not a clean separation of any single mechanism.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with `[::2]` subsampling to 12.5 fps, preserving the donor’s original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α -blending of upper-body joints ($\alpha=0.88$; full joint-set definitions in the artifact). The matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the probe response co-occurs with donor origin, we construct frozen probe systems: **UNSEEN-PURE-v1** retrieves from the 640 other test poses;

CROSSFIT-PURE-A/B split queries by hash into two folds; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ± 0.1 Jaccard of each UNSEEN donor (622/641 admit one). SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry; these are constructed probes, not deployable generators (construction rules in the artifact).

3.3 Evaluator checkpoint family

The primary inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and six *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture on a common frozen partition (primary seeds 101–606; eight extension seeds 707–1405 use the identical protocol). The original bundle’s `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈ 102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.3.1, and intermediate checkpoints are not published. We use two reconstruction protocols — *paper-derived* (14 runs: 6 primary seeds 101–606, 8 extension seeds 707–1405; legacy script, 300 epochs, NLL selection) and *near-faithful* (6 runs: 4 config-faithful + 2 step-faithful; decoded-BLEU early stopping) — sharing the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), `[:: 2]` subsampling, Adam (1e-3, wd .001), batch size 256, and plateau $\times 0.8$ LR schedule (full provenance in SI). In total 70 from-scratch training runs across 13 families (registry in SI Sup. D); 41 were beam-3 decoded on PHX-public under the canonical §3.2 donor registry, corresponding to 40 unique weight binaries — one config-faithful seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run, identical per-step losses and best-checkpoint SHA-256). Of the 41 decoded training runs, five are degenerate (three $\alpha=1.0$ distillation students with empty hypotheses, two smallest ladder fractions with BLEU ≈ 0) and are excluded, so the non-replication result rests on 36 non-degenerate decoded training runs (all with strictly negative gaps). Table 7 gives the full accounting.

What the 70 runs are—and are not.

The 70 runs are *not* 70 same-recipe replication attempts. Table 3 maps each family to the public source it draws on, the known deviation from the original training, the number of runs, the checkpoint-selection rule, and the resulting dev BLEU-4 range. Only the 2 step-faithful runs and the 4 config-faithful runs approach a faithful reconstruction of the released recipe; the remaining 60 runs are diagnostic, sensitivity, holdout, or competence-extrapolation analyses. The public-recipe non-reproduction conclusion rests on the 6 recipe-faithful runs, while the remaining families serve as triangulating evidence that no alternative construction we tested reproduces the reversal.

Training-pool readout (descriptive observation).

We define *training-pool readout* as the in-sample autoregressive free-decode BLEU and exact-match (EM) rate on the full 7,060-item training set under beam-3 decoding—not teacher-forced accuracy. The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match, while retaining dev/test

Table 3: Public-artifact provenance of each reconstruction family. “Dev BLEU-4” is the uniform full-pool beam-3 value (§3.4); the released evaluator scores 13.38. “Selection” is the checkpoint-selection rule. The 14 paper-derived runs use NLL selection on a legacy script; the config-faithful runs originally misread `validation_freq=14` as epochs before the step-faithful correction. Only recipe-faithful families support the public-recipe non-reproduction claim; the rest are diagnostic.

Family	Public source used	Known deviation	Runs	Selection	Dev BLEU-4
Paper-derived (primary+ext.)	config.yaml + slt architecture	legacy script; epochs; NLL selection	300 14	NLL on dev	6.5–9.4
Config-faithful	config.yaml (decoded-BLEU)	<code>validation_freq=14</code> misread as epochs initially	4	decoded BLEU	8.3–9.8
Step-faithful	corrected step counting	fixed <code>validation_freq</code> interpretation	2	decoded BLEU	9.1–9.8
Confirmation	recipe-faithful protocol	independent re-run for reproducibility	2	decoded BLEU	9.0–9.5
Long-schedule	config.yaml (epochs=3000)	extended training budget	2	decoded BLEU	≤10.02
Rescue sweep	hyperparameter grid	dropout/wd/lr/batch variations; none passed gate	20	decoded BLEU	≤10.02
Distillation	released checkpoint as teacher	student-teacher KL; inherits released weights	KL; 15	decoded BLEU	7.0–9.0
Ladder / large-arch	various	data-subset fractions; architecture scaling	11	various	≤9.8
Total			70		

performance of 13.38/12.78. The constructible family’s readout is far lower (reconstructions: 7.2–9.3 BLEU, 1.5–2.3% EM; distillation: 6.6–8.9 BLEU, 1.3–1.9% EM); no constructible checkpoint approaches both simultaneously. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$ vs. reconstruction family: 0.93–1.12) quantifies how much the model reads out above its in-domain performance. We report these as continuous descriptive observations: the released evaluator was observed first, the family range afterwards, and any threshold is post-hoc.

3.4 Reconstruction sanity checks

Because all 70 training runs share the same codebase, data pipeline, and evaluation protocol, we performed three checks to rule out a systematic implementation defect:

- all-sample overfit test.** A reconstruction model trained on 50 and 100 sequences overfits to free-decode BLEU 100.00 ($\text{NLL} \rightarrow 0.009$), ruling out defects in loading, masking, or decoding (`results/overfit_test.json`).
- inference regression.** Released and reconstruction checkpoints load through the identical pipeline with byte-identical vocab files (SHA-256 confirmed); the released checkpoint reproduces canonical REC 12.78 exactly.

her-forced vs. free-decode. Table 4 shows the released evaluator achieves 96.5% teacher-forced train token accuracy and 78.8 free-decode train BLEU, while reconstructions achieve 55–60% and 7.2–9.3. The gap is present in both teacher-forced and free-decode metrics on both train and dev sets.

Table 4: Teacher-forced and free-decode diagnostic statistics for the released evaluator and the fourteen reconstruction seeds (full table in `results/checkpoint_stats.json`). Teacher-forced values are on 500 random sampled items (seed 42, sampled without replacement; the sampling SD across 5 re-samples is ≤ 0.3 pp for token accuracy and ≤ 0.01 for NLL/tok); free-decode values are full-pool (7,060 train / 515 dev) under the uniform beam-3 protocol (§3.4). sacreBLEU **signature:** BLEU|c|n:4|e:no|tok:13a|s:exp|v:2.5.1.

Checkpoint	Train tok. acc.	Train NLL/tok	Train free-decode BLEU	Dev NLL/tok	Dev free-decode BLEU
Released	0.965	0.158	78.82	3.236	13.8
Reconstructions (range)	0.550–0.596	1.72–1.92	7.20–9.25	2.63–2.68	6.49–9.3
Reconstructions (mean)	0.572	1.82	8.35	2.66	8.0

3.5 Discovery timeline and pre-registration status

The +10.24 probe response was first observed on 2026-07-24 (canonical materialization; earlier pre-exclusion gap +11.01 on 2026-07-22). The six primary reconstruction seeds (101–606) were specified before their results were inspected; the eight extension seeds (707–1405) were added post-hoc. The Jaccard retriever, PURE protocol, and 641-query set were locked before reconstruction training began, after exploring the probes in Table 8. No external pre-registration exists; the six primary seeds constitute the only pre-specified confirmatory evidence. Because the Jaccard pure-copy probe was selected as the maximal-gap probe post-hoc, its CI is descriptive, not confirmatory. Full provenance in SI Sup. C.

3.6 Metrics and uncertainty

The recorded test poses (REC) are decoded by each BT recognizer; we avoid the abbreviation “ground truth” because recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale Post (2018) (signature: BLEU|c|n:4|e:no|tok:13a|s:exp|v:2.5.1). BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020), and WER (jiwer 3.1.0) provide cross-metric consistency checks (not independent semantic validation, as all share decoded outputs). BT likelihood and pose-DTWp provide non-decoded controls (SI Sup. O).

Uncertainty is quantified with donor-cluster bootstrap (primary; 10,000 resamples) and item-level bootstrap (SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. A descriptive seed-level 95% Student- t prediction interval is reported for the legacy-implementation family only; its interpretation is “given a

future random seed under this identical implementation,” not a population-level claim. The six primary seeds give PI $[-2.48, -0.28]$; all 14 leave-one-seed-out folds keep the interval below zero (SI Sup. B). The headline gap is a **post-selection descriptive estimate** (§4.1); all other contrasts are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

3.7 Terminology: run, checkpoint, unique binary, family, evaluator

Throughout, *run* denotes a single training execution; *checkpoint* denotes the selected model weights from that run; *unique weight binary* counts distinct SHA-256 hashes of checkpoint files (the cf.202 and ls.202 runs produced byte-identical weights from the same wandb run); *family* denotes a group of runs sharing a training protocol; and *evaluator* denotes a checkpoint used to decode fixed outputs. The 14 paper-derived reconstruction runs use NLL selection on the legacy script; the 4 released-config runs use decoded-BLEU early stopping. Both share the documented architecture and hyperparameters (§3).

Custom-terminology registry.

Table 5 maps each author-defined term to its estimand, dataset, and evidence grade, following the reviewer’s request. Evidence grade: D = descriptive (no formal error control); all custom quantities in this paper are descriptive.

Table 5: Custom-terminology registry. Each term maps to its estimand, the dataset on which it is computed, and its evidence grade (all descriptive).

Term	Estimand	Dataset	Grade
Training-pool read-out	In-sample autoregressive free-decode BLEU/EM	Full 7,060-item training set	D
PURE-REC gap	Corpus sacreBLEU-4 difference (PURE – REC)	641-item PHX-public test	D (post-selection)
Operational tolerance gate	One-sided: dev BLEU-4 ≥ 12.38 and dev WER ≤ 86.37	515-item dev set	D

Claim hierarchy.

We classify every claim in this paper into four evidence tiers:

- *Descriptive (D)*: observed pattern with no formal error control. Most findings in this audit are descriptive.
- *Sensitivity (S)*: the result is qualified by an explicit post-hoc condition (e.g., single-annotator, matched subset, post-selection).
- *Confirmatory (C)*: pre-specified hypothesis with formal error control. Only the six primary reconstruction seeds’ non-reproduction direction qualifies.

- *Not identified (NI)*: the claim cannot be made from available data. The causal checkpoint-identity effect is NI because competence is confounded.

Table 6: Claim hierarchy. Each key claim mapped to its evidence tier and supporting evidence.

Claim	Tier	Evidence
Public artifacts do not reproduce released evaluator competence	D	20 recipe-conditioned runs, all dev BLEU < 10.02
Probe response does not reproduce across weaker checkpoints	D	36/36 non-degenerate gaps negative
Probe response reflects a causal checkpoint-identity effect	NI	Competence confound: 6.5–9.8 vs 13.38
Headline gap +10.24 is post-selection	S	Max-gap probe among 8 explored; CI descriptive
Reference replacement attenuates the gap (90.5%)	S	Single-annotator, 461-item, 4 confounded changes
Origin effect persists under matched pools (+9.56)	S	148 items, SMD<0.1; association not causal
Probe is adversarial lexical-coupling stress test	D	Source = reference in PHX-public; retriever uses source Jaccard

Experimental stratification.

We classify every checkpoint family into four evidential tiers:

- *Primary (near-faithful reconstruction)*: 4 config-faithful + 2 step-faithful = 6 runs. These approach the released training recipe most closely (step-correct validation frequency, decoded-BLEU selection). Their results are the primary evidence for the reconstruction-failure claim.
- *Secondary (legacy-implementation replication)*: 14 paper-derived seeds (6 primary + 8 extension; legacy script, 300 epochs, NLL selection). These share a known deviation from the released protocol and serve as supplementary stability evidence.
- *Diagnostic (post-hoc sensitivity)*: 2 confirmation + 20 rescue + 15 distillation + 4 ladder + 2 long-schedule + 4 large-arch = 47 runs. These probe mechanism and competence extrapolation.
- *Holdout controls*: 2 cross-fit + 1 BT-retrained = 3 runs, evaluated on holdout splits only.

Main conclusions rest on the 6 primary runs; the 14 secondary runs provide stability evidence; diagnostic runs serve as triangulating context and must not be cumulated with primary to inflate the appearance of independent replication.

Table 7 gives the full accounting, generated from a single machine-readable registry (`results/accounting_table.json`).

Table 7: Checkpoint accounting, auto-generated from `results/accounting_table.json` (single source of truth). “Runs” = from-scratch training executions; “Decoded” = beam-3 decoded on the canonical PHX-public gap panel; “Unique” = distinct SHA-256 weight binaries; “Non-deg.” = non-degenerate (BLEU > 0, excluding 3 $\alpha=1.0$ distillation students with empty output and 2 smallest ladder fractions). The primary tier (6 near-faithful) supports the reconstruction-failure claim; secondary (14 legacy) adds stability; diagnostic and holdout are mechanism exploration.

Tier / Family	Runs	Decoded	Unique	Non-deg.
Primary	6	6	6	6
Config-faithful	4	4	4	4
Step-faithful	2	2	2	2
Secondary	14	14	14	14
Paper-derived (primary seeds)	6	6	6	6
Paper-derived (extension seeds)	8	8	8	8
Diagnostic	47	21	20*	16
Confirmation	2	2	2	2
Rescue	12	1	1	1
(dropout/wd variants)				
Rescue (lr variants)	8	0	0	0
Distillation	15	13	13	10
Ladder (data fractions)	4	4	4	2
Long-schedule	2	1	0*	1
Large-arch	4	0	0	0
Holdout controls	3	0	0	0
Total trained	70	41	40	36

*One config-faithful seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run); it counts once in the unique-binary column. The released evaluator (1 checkpoint) is not a training run and is not included above. Eleven released-weight fine-tune variants (Gaussian noise, local perturbation) are not from-scratch training and are not counted here.

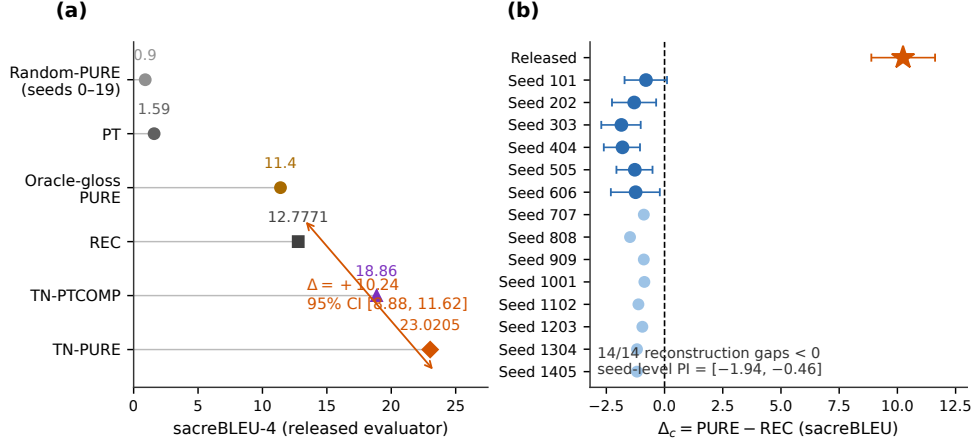


Fig. 1: Headline reversal under the released evaluator and its absence across recipe reconstructions. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, 95% CI [8.87, 11.64]), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE–REC gaps: released evaluator (+10.24) vs. fourteen recipe reconstructions, all negative (donor-cluster bootstrap 95% CIs; seed-level 95% PI [−1.94, −0.46]). All reconstructions are less competent than the released evaluator; the figure demonstrates public-pipeline non-replication, not an isolated causal checkpoint effect.

4 Results

4.1 Adversarial probe response and conditioning controls

Under the released SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses—a gap of +10.24 BLEU points (95% CI [+8.87, +11.64]; 10,000 donor-cluster resamples; item-level bootstrap gives the nearly identical [8.88, 11.62]). This is the expected output of a deliberately adversarial lexical-coupling stress test (§3.2): the retriever selects donors by surface vocabulary overlap with the source, which is identical to the scoring reference, so the gap measures the evaluator’s susceptibility to this specific shortcut rather than general retrieval quality or signing adequacy. A deployment-realistic noisy-gloss retrieval pipeline scores only −0.13 relative to REC under the same evaluator (§4.5), confirming that the gap is specific to the constructed adversarial probe.

Controls localized the observation to source-conditioned donor retrieval (SI Table S22): pure donor copy (23.02) exceeded PT-composed (18.86), oracle-gloss (11.4–11.8, a leakage diagnostic), PT baseline (1.59), and 20-seed random-donor baselines (0.90/0.77). Source-text Jaccard was the strongest retriever (vs. LaBSE 19.2, multi-SBERT 16.5). Non-decoded controls do not reproduce the gap: teacher-forced

NLL gives REC 3.060 vs PURE 3.479, and pose-DTW is near zero for all non-REC conditions. Complementary text metrics (BLEU-1, chrF, ROUGE-L, WER, BERTScore) agree on direction (SI Table S22); decoding-parameter sweeps (beam 1/3/5, length penalty, max length) leave the gap at +9.9 to +10.8 (SI Sup. F).

Probe multiplicity and post-selection status of the headline gap.

Before freezing the headline probe, we explored four retrieval strategies (source-text Jaccard, LaBSE, multi-SBERT, oracle gloss) crossed with three construction variants (pure copy, PT-composed, oracle-gloss-composed) on the released evaluator alone. The full design matrix comprises 12 cells; 8 were explored and 4 were not (Table 8 and `results/probe_multiplicity.json`). The unexplored cells are: LaBSE×PT-composed and multi-SBERT×PT-composed (deemed uninformative after Jaccard pure-copy dominated), and two oracle-gloss-composed cells (subsumed by oracle-gloss×PT-composed or definitionally inapplicable). The headline Jaccard pure-copy probe was selected as the *maximal-gap* probe; its 95% CI [+8.87, +11.64] is therefore **post-selection descriptive**. A Bonferroni correction over the family of 8 explored probes ($\alpha=0.05/8=0.00625$, $z^*\approx 2.66$) would still exclude zero (the gap’s z -statistic is ≈ 14.7), but we label the result descriptive because the family was defined after observing the gaps. The six primary reconstruction seeds (§3.5) were pre-specified and provide the only confirmatory evidence (for the non-reproduction direction, not for the gap magnitude).

Table 8: Complete retrieval × construction design matrix explored on the released evaluator. Scores are corpus-level sacreBLEU-4 (0–100) on the canonical 641-item PHX-public set (REC=12.78). The headline probe (bold) was selected as the maximal-gap probe among 8 explored cells; 4 cells were not explored (reasons in `results/probe_multiplicity.json`). The headline CI is post-selection descriptive.

Retriever	Construction	Explored	Score	Gap vs. REC
Source-text Jaccard	Pure donor copy	✓	23.02	+10.24
Source-text Jaccard	PT-composed	✓	18.86	+6.08
Source-text Jaccard	Oracle-gloss-comp.	—	—	not applicable
LaBSE	Pure donor copy	✓	19.2	+6.42
LaBSE	PT-composed	—	—	not explored*
multi-SBERT	Pure donor copy	✓	16.5	+3.72
multi-SBERT	PT-composed	—	—	not explored*
Oracle gloss	Pure donor copy	✓	11.4	−1.38
Oracle gloss	PT-composed	✓	11.8	−0.98
Random donor	Pure donor copy	✓	0.90	−11.88
Random donor	PT-composed	✓	0.77	−12.01

*Deemed uninformative after Jaccard pure-copy dominated; composition with weaker retrievers cannot exceed the Jaccard gap.

The probe reversal motivates the prior question: can the public recipe reproduce the released evaluator’s competence at all?

4.2 RQ1: Failed public-artifact reconstruction—what the released evaluator has that reconstructions lack

The non-reproduction in RQ2 raises the prior question of RQ1: can the public recipe reproduce the released evaluator’s competence? We approach it through three descriptive comparisons (Fig. 2b,c; Table 9; full version in SI Sup. D), reporting them as descriptive triangulation, not three independent tests.

Table 9: Summary of the three primary descriptive comparisons. The released evaluator sits far outside the constructible family range on all three views. Per-view family composition in SI Sup. D.

Comparison	Family n	Released evaluator	Family range	Released – family m
(i) Gap–competence	42	+10.24 at dev 13.38	$[-2.01, -0.21]$ (36 non-degenerate runs)	+10.
(ii) Train-pool read- out BLEU	27	78.8 (70.7% EM)	6.6–9.3 (1–2% EM)	+69.
(iii) Competence gate	52	13.38	6.5–9.8 (uniform; training-log max 10.02)	+3.

Readout with generalization: the distinctive combination.

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match while retaining dev/test performance of 13.38/12.78. The constructible family stays far below: reconstructions span train BLEU 7.2–9.3 at dev BLEU 6.5–9.4, and distillation students 6.6–8.9 at dev 7.0–9.0. Across 31 checkpoints with uniform full-pool readouts (30 constructible plus released), training-pool readout tracks decoded competence (Spearman $\rho=0.916$, $n=31$) and is uncorrelated with the overfit indicator ($\rho=0.040$) or the PURE–REC gap ($\rho=-0.082$; SI Sup. D). The released evaluator’s readout-with-generalization combination is not explained by any within-family overfit–readout relationship.

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact: five controls (SI Sup. M) rule out four shortcut classes — teacher forcing, output–reference misalignment (label permutation collapses BLEU from 78.6 to ~ 1.0), pose-ignorant ID/cache lookup (input-side pose permutation collapses BLEU to <1 ; a same-signer held-out control reaches only 12.19 BLEU / 3.2% EM), and narrow-text memorization. The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure.

Distillation and competence extrapolation.

Fifteen distillation students ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, three seeds each) all have negative gaps and do not transfer the readout pattern (SI Table S26). Competence

extrapolation (OLS/GP), per-item gap decomposition, and Gaussian-weight-noise perturbation are reported in SI Sup. K and F; neither resolves the identification problem inside the unobserved (9.8, 13.38) interval. The reversal is locally robust near the released checkpoint (Gaussian noise $\sigma \leq 3 \times 10^{-3}$ preserves it within ± 0.75 ; 9 released-weight fine-tunes all pass the competence gate), but these probes test the released checkpoint’s vicinity, not an independent construction.

Synthesis.

Across the three descriptive comparisons, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78). No reconstruction or distilled student reproduces either the +10.24 gap or this combination.

4.3 RQ2: Multi-checkpoint reproduction of the stress-test ordering

TN-PURE-v1 yields an original-evaluator gap of +10.24 BLEU points (95% CI [+8.87, +11.64]) and TN-PTCOMP-v1 +6.09 [+4.76, +7.41]; across all fourteen reconstructions both gaps are negative (PURE range $[-1.85, -0.80]$; seed-level 95% prediction interval $[-1.94, -0.46]$, mean -1.20 , SD 0.33, 0/14 positive; the six pre-specified primary seeds alone give $[-2.48, -0.28]$ and the eight extension seeds $[-1.61, -0.54]$, with the interval below zero in all fourteen leave-one-seed-out folds), and for the six primary runs original-minus-run difference-in-differences estimates range from +10.89 to +11.99, every interval excluding zero.

4.3.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 10), and all evaluators are decoded with the canonical §3.2 donor registry (exact-normalized-text exclusion applied).

Competence is assessed on the released 515-sequence development subset, never on PHX-public. The released evaluator obtains dev BLEU-4 13.38; under the uniform full-pool beam-3 free-decode protocol the 14 paper-derived reconstructions obtain dev BLEU-4 6.5–9.4, and the 6 near-faithful runs obtain 8.3–9.8. Their selected validation NLL (2.641–2.712) is lower than the released evaluator’s teacher-forced dev NLL (3.235), so likelihood and decoded recognition competence are not interchangeable. Consequently, competence and unavailable original-training details remain confounded with evaluator identity; this is a descriptive observation, not a causal checkpoint-identification result.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both dev BLEU-4 ≥ 12.38 and dev WER ≤ 86.37 (one-sided tolerance around released 13.38 / 83.37). None of the 52 gate-eligible non-distillation training runs passes the gate. The 9 released-weight fine-tunes (SI Sup. F) all pass (dev BLEU-4 12.92–13.18), but they perturb the released weights

Table 10: Canonical fixed-output evaluation on the released 641-pose PHX-public subset (released evaluator and six primary reconstruction seeds; eight extension seeds in SI Table S27). PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1. Gaps are relative to local REC.

Evaluator	REC	PURE	Gap	COMP	Gap
Released	12.78	23.02	+10.24	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16
Extension 707–1405 (range)	see SI Table S27	−1.48 to	−0.87	−1.03 to	−0.44

rather than constructing from the recipe. The rescue search’s best variant (weight-decay 0, seed 202) yields gap -1.59 . The gate conclusion is threshold-insensitive across a $\pm[0.5, 2.0]$ BLEU $\times \pm[1.5, 4.5]$ WER grid. Full details in SI Sup. D–E.

What the released training log establishes.

The bundle’s `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 2a), while all constructible reconstructions plateau at 8–10 by epoch 50–100; the training dynamic behind this slope is not explained by public artifacts. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at $k=8$: REC 8.75, PURE 11.69, gap +2.94; SI Sup. F).

Cross-metric consistency (SI Table S22).

Under the original evaluator, the PURE–REC gap is positive and large for all decoded-text metrics (BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER +13.90), while teacher-forced NLL is more negative for PURE (-0.419 sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under all six primary reconstructions the gap is negative across every metric. The probe response is therefore a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4. Full table in SI Table S22.

Rank stability (SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean $\tau_b=0.44$, $\rho=0.57$).

Rank stability (see SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean $\tau_b=0.44$, $\rho=0.57$).

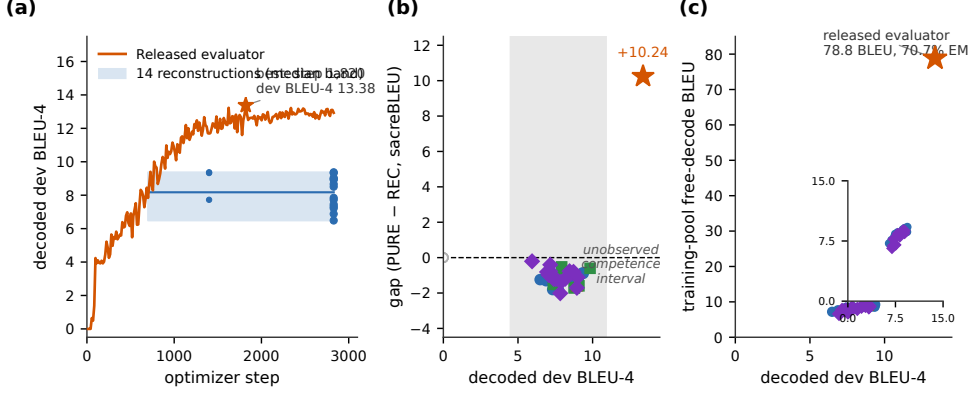


Fig. 2: Decoded competence and training-pool readout separate the released evaluator from every constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (archived validation log, 202 logged points); fourteen reconstructions form a median band (blue), plus uniformly decoded epoch-25/50 points. (b) Canonical PURE-REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), config-faithful/step-faithful/confirmation/long-schedule (squares), rescue (triangle), distillation students (diamonds), released evaluator (star); the gray band marks the observed 6.5–9.8 range (uniform protocol) and the (9.8, 13.38) interval is unobserved by recipe-constructed checkpoints (the nine released-weight fine-tunes occupy 12.92–13.18 but are weight perturbations); degenerate checkpoints are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses) vs. decoded dev BLEU-4: every constructible checkpoint reads out 6.6–9.3 BLEU, while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38. Families are descriptive; checkpoints are not IID replicates.

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train-test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap

is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a paired attenuation of 9.08 BLEU points (90.5%, bootstrap CI [7.44, 10.86] on the absolute attenuation and [0.81, 1.01] on the attenuation ratio; SI Table S31; SI Fig. S5; per-item paired view in SI Fig. S2). The interval for the residual gap includes zero (paired item CI [−0.13, 1.99]). This 90.5% is a *single-annotator, 461-item, post-hoc point estimate*; it cannot isolate any single mechanism because reference replacement simultaneously changes (i) reference content, (ii) expression style/translationese, (iii) BLEU floor level, and (iv) the lexical coupling between retrieval selection and scoring reference. Teacher-forced BT likelihood does not reproduce the probe response under either reference (REC NLL 3.060 vs PURE 3.479 on original refs; 3.41 vs 3.71 on human refs) Jiang et al. (2025).

Scoring against both references jointly leaves the gap at +11.50, *larger* than original: adding the original reference injects a second lexically coupled target. The probe response is thus specific to reference frames containing the original speech-transcribed register. This is a descriptive contrast, not a causal attribution. Under all six reconstruction checkpoints the human-reference gap is negative (−0.15 to −0.65), and the asymmetry is consistent across metrics (chrF 82.2%, BLEU-1 84.0%; SI Table S23). The asymmetric PURE-vs-REC drop ($1.31 \times \text{BLEU-4}$) is numerically inconsistent with simple symmetric floor compression (SI Table S23), though BLEU’s non-linearity precludes a definitive test. The human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so semantic adequacy of TN-PURE-v1 is left unassessed.

4.4 Donor-origin contrast

If the released-evaluator probe response requires exposure to training-pool donors, retrieval restricted to unseen poses should remove or invert it: under the original evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU vs. REC 12.78 (gap −4.27), a donor-pool substitution estimand of +14.51 BLEU points (SI Table S30). This is a *donor-pool substitution effect*, not a pure exposure effect: SEEN and UNSEEN differ simultaneously in pool size (7,060 vs 640), median donor-reference Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure.

Matched donor-pool analysis (reviewer-requested $\text{SMD} < 0.1$ control).

To separate the donor-similarity confound from the pool-origin effect, we computed a per-item matched subset restricting to items where the seen-donor and unseen-donor Jaccard similarities differ by at most 0.05. This yields 148 items with $\text{SMD} = 0.065$ on Jaccard (vs. +0.554 on the full 641). On this matched subset, the origin effect is +9.56 BLEU [+7.47, +11.76]—reduced from the unmatched +14.51 but still large and with CI excluding zero. Approximately 34% of the origin effect is attributable to donor-similarity differences; the remaining 66% persists under similarity matching. This residual is an association, not an identified training-exposure effect: the matched analysis controls for donor-reference Jaccard but not for pool size (7,060 vs 640) or the diversity of available donors. Full results in `results/matched_donor_pool.json`; additional balance checks in SI Sup. E.

Balanced factorial and path decomposition.

The 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity; SMD +0.4 within strata) estimates the pool-origin main effect at +2.08 [+1.56, +2.57], similarity at +6.96 [+5.93, +8.08], and interaction at +3.70 [+2.63, +4.79] (Holm $p=0.0003$); across reconstructions the seen main effect shrinks to +0.51. Because seen donors remain more similar than unseen within strata, this is a provenance contrast, not an identified training-exposure effect. The matched-subset analysis above provides a tighter estimate; the factorial is retained for its 2×2 decomposition.

4.5 Competitive-system context

Because the audit’s adversarial probe is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: the “Ground Truth” row reproduces our canonical REC numbers exactly (BLEU-4 12.78, Walsh et al. (2025); SI Table S25). The retrieval-based winner USTC-MoE sits roughly even with REC on lexical BT metrics (BLEU-4 −0.72; SI Table S25). We re-decoded two realistic retrieval outputs under all available evaluators (SI Sup. P): the deployment-realistic noisy-gloss retrieval pipeline decodes at 12.65 under the released evaluator (gap −0.13 vs. REC), and our rebuild of the winner’s fixed outputs from its released intermediates decodes at 2.42 (gap −9.94; a failed reproduction, not representative of the original entry). Under all 36 non-degenerate reconstructed evaluators, both systems give negative gaps (noisy-gloss range [−1.94, −0.07]; USTC-MoE range [−8.30, −4.28]). Neither realistic system recovers a retrieval-favoring ordering under any evaluator. The +10.24 gap is specific to the constructed adversarial probe, and we do not claim that real competition rankings would change across checkpoints. Archiving fixed per-team outputs alongside public scores is the most direct suggestion we can offer for enabling rank-stability audits.

Per-item gap decomposition (experiment C, SI Sup. K).

An OLS regression ($R^2 = 0.33$, $n = 641$) attributes the largest share of per-item gap variance to high-overlap-train count, template density, and donor–reference Levenshtein distance; signer identity contributes $\leq 1\%$. Full model, coefficient table, and figure in SI Sup. K.

5 Discussion

The audit yields three findings organized around the question of whether the publicly documented recipe is sufficient to reproduce the released SLRTP2025 BT evaluator and its behavior on a constructed replay probe.

RQ1—Failed public-artifact reconstruction. Using our disclosed reconstruction implementation and the publicly released artifacts, none of the 6 near-faithful runs (4 config-faithful, 2 step-faithful) or 14 legacy-implementation seeds reached the released evaluator’s competence (dev BLEU-4 6.5–9.8 vs. 13.38), training-pool readout (7.2–9.3 vs. 78.8), or teacher-forced token accuracy (55–60% vs. 96.5%; Table 4). Three descriptive comparisons (Table 9) all place the released evaluator far outside any constructible range. The source of the gap remains unidentified.

RQ2—Descriptive non-reproduction under competence-unmatched reconstructions. Because every constructible checkpoint is less competent than the released evaluator (dev BLEU-4 6.5–9.8 vs. 13.38), the non-reproduction of the adversarial probe response (all 36 non-degenerate gaps negative, range $[-2.01, -0.21]$) *cannot distinguish* whether it reflects the competence gap, an unrecovered training mechanism, or their interaction. This is a descriptive observation about our public-artifact reconstruction attempt, not a causal checkpoint-identity claim. Without a competence-matched independent implementation—which we could not obtain (the released repo is eval-only; our trained checkpoints were lost in a host compromise)—the identification problem is unresolved. Cross-evaluator re-decoding of a realistic noisy-gloss retrieval output yields a released-evaluator gap of -0.13 and 36/36 negative gaps, confirming the probe response is specific to the constructed adversarial design.

Because all constructible checkpoints are less competent than the released evaluator (decoded family max dev BLEU-4 9.8, training-log max 10.02, vs. 13.38), this non-reproduction is scoped to the public recipe and does not constitute a causal checkpoint-identity claim. Two local-robustness checks on the released checkpoint (Gaussian weight noise and fine-tuning from released weights, SI Sup. F) confirm the reversal is locally robust but probe only the vicinity of the released weights. Competence-extrapolation and per-item decomposition analyses (SI Sup. K) are exploratory and do not resolve the identification problem. Our rebuilt USTC-MoE output (SI Table S33) is pipeline-faithful but not score-faithful (2.42 vs. official 12.06) and should not be taken as representative of the original competition entry. The noisy-gloss result provides the primary realistic-system evidence.

RQ3—Reference-frame sensitivity and donor-origin coupling. On the matched 461-item confidence=1 subset of Czehmann et al.’s Czehmann et al. (2026) human back-translations, reference replacement is associated with a gap reduction to $+0.95$ from $+10.03$ —a single-annotator, 461-item, post-hoc point estimate of 90.5% attenuation. This estimate cannot isolate any single mechanism because reference replacement simultaneously changes (i) reference content (speech transcription vs. sign-conditioned back-translation), (ii) expression style and translationese, (iii) BLEU floor level (REC drops 7.59, PURE drops 16.68), and (iv) the lexical coupling between retrieval selection and the scoring reference. The asymmetric PURE-vs-REC drop ($1.31 \times$ BLEU-4) is numerically inconsistent with simple symmetric floor compression, but BLEU’s non-linearity and the coupling structure preclude a definitive floor-effect test. The matched donor-pool analysis (§4.4) shows the origin effect persists at $+9.56$ even under $\text{SMD} < 0.1$ Jaccard matching (148 items), confirming that approximately two-thirds of the origin effect is not explained by donor-similarity confounding. The donor-pool substitution reverses the gap sign (UNSEEN-PURE-v1 scores 8.51 vs. REC 12.78, gap -4.27); both the matched and unmatched origin effects remain associations, not identified training-exposure effects, because pool size and donor diversity are not equalized.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-recipe reproducibility: improved BT scores need not imply target conditioning Hong

and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal — teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479) — and whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025).

Value of the audit infrastructure despite the competence confound.

The competence-matched condition is unmet, so RQ2’s non-reproduction cannot be attributed to a causal checkpoint effect. This constrains what the audit *infers* but not what it *provides*: (i) the provenance table (Table 2) and three-way data reconciliation establish the released materialization differs from the official split by 41 IDs; (ii) the checkpoint registry with SHA-256 deduplication (SI Sup. D) prevents double-counting; (iii) the unified-inference regression test (§3.4) isolates code-path defects from recipe insufficiency; (iv) the leakage controls (SI Sup. M) are extensible to any learned evaluator; (v) the single canonical materialization with four-layer reproduction (§3) is a concrete model for reproducible benchmarking infrastructure. These contributions are transferable to future learned-evaluator audits.

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: a released evaluator is a research artifact, not an infrastructure component. The released `config.yaml` alone was insufficient for 70 reconstruction runs across 13 families to reach competence parity, and the readout pattern could not be localized to a training stage because intermediate checkpoints are unpublished. Benchmarks should release (i) the exact training command and seeds, (ii) data pipeline scripts beyond the manifest, (iii) a SHA-256 manifest of training tensors, (iv) intermediate checkpoints at least every 100 epochs, and (v) the fixed per-team pose outputs alongside the public scores; the current release provides only the Progressive Transformer baseline output, so rank-stability audits across independently trained evaluators cannot be reproduced by third parties on real competition systems.

Lesson 2: on template-dense benchmarks, retrieval entries may benefit from donor-exclusion disclosure. A donor-exclusion sweep is cheap and exposes the retrieval advantage on PHX-public’s template-dense register; retrieval-based leaderboard entries *may* voluntarily attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results as a transparency practice on template-dense benchmarks. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$; reconstructions: 0.93–1.12) and input-side pose permutation with output-equivariance verification (SI Sup. M) are two descriptive probes we found informative in this case study; whether they generalize to other benchmarks is untested.

Lesson 3: reference-coupling artifacts warrant a supplementary reference check. On PHX-public, the 90.5% attenuation under Czehmann et al. Czehmann et al. (2026) back-translations is a single-annotator reference-frame case study showing that the gap is sensitive to the reference frame; it should not be generalized beyond this signer’s

interpretation. Benchmarks on similarly template-dense corpora could adopt human back-translations as a supplementary reference set, follow the matched-subset protocol, and flag systems whose gap shifts between the two; the cost is one reference file and a paired bootstrap. Whether this generalizes beyond PHX-public’s lexical-coupling structure is untested.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train-test-overlap issues Czehmann et al. (2026); Alkain et al. (2026). Per-team pose outputs are not archived; our cross-evaluator studies decode at or below REC under all 36 non-degenerate evaluators. All 70 constructible checkpoints (Table 7) are less competent than the released evaluator (max dev BLEU-4 9.8 vs. 13.38). Without a competence-matched independent implementation, the causal checkpoint-identity question is unresolved. The human back-translations are a single-annotator resource without second-annotator adjudication; the 90.5% attenuation is conditional on one signer’s interpretation and should not be taken as a population parameter (qualitative direction is stable across confidence tiers; SI Table S31). The probe response is a decoded-BT-text-metric effect, not a sign-language quality judgment; without target-conditioned DGS signer evaluation, semantic adequacy of the donor motion is unassessed.

6 Conclusion

This audit reports a failed public-artifact reconstruction attempt of the released SLRTP2025 BT evaluator and a descriptive non-reproduction of a constructed adversarial probe response.

For RQ1, none of our 6 near-faithful or 14 legacy-implementation reconstructions reached the released evaluator’s competence; the source of the gap remains unidentified. For RQ2, all 36 non-degenerate constructible checkpoints give negative probe gaps, but because every constructible checkpoint is less competent than the released evaluator (dev BLEU-4 6.5–9.8 vs. 13.38), we cannot distinguish whether this reflects the competence gap, an unrecovered training mechanism, or their interaction. For RQ3, reference replacement is associated with a gap reduction to +0.95 (interval includes zero) in a single-annotator, 461-item case study that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects.

The scope is bounded: no causal checkpoint effect, no identified memorization mechanism, and no general leaderboard-rank instability is established. We frame the replay probe as an adversarial lexical-coupling stress test throughout. The audit’s lasting contributions are the provenance, checkpoint-registry, and evaluator-audit infrastructure, transferable to future learned-evaluator audits.

Statements and Declarations

Funding.

The authors declare that no funds, grants, or other support were received.

Conflict of interest.

The authors declare no competing interests.

Data governance and ethics.

Although all datasets are publicly released, pose tensors retain signer identity, biometric features, and signing style. We address four governance dimensions. (i) *Consent scope*: we conduct this research under the published licences of PHOENIX-2014T and SLRTP2025; however, the original consent forms are not publicly available, so we cannot independently confirm that the informants’ consent covers derivative replay-probe analysis. No new human-subjects data was collected. (ii) *Licensing*: PHOENIX-2014T is distributed under its own academic research licence (the version specified in the distributed copy; the official landing page does not display a Creative Commons version); the Czehmann et al. Czehmann et al. (2026) back-translations are CC BY-NC-SA 4.0; CSL-Daily and SLRTP2025 carry academic-use licences; the licence compatibility matrix is documented in the artifact. (iii) *Identity re-identification risk*: we release per-item decoded text (low re-identification risk) but do *not* release per-signer raw poses; the artifact ships only aggregated metrics, decoded hypotheses, and analysis scripts. (iv) *Deaf community involvement*: no Deaf/DGS stakeholders were involved in formulating the research questions or interpreting the results of this audit; we identify this as a limitation. The Czehmann et al. back-translations we build upon were produced by a Deaf fluent L2 DGS signer.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the artifact at <https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md>; a persistent DOI-bearing release will be deposited upon acceptance. The supplementary material (**supplementary.pdf**) contains all SI appendices. Upstream datasets are used under their published licences (PHOENIX-2014T under its academic licence; the Czehmann et al. human back-translations under CC BY-NC-SA 4.0; CSL-Daily and SLRTP2025 under their academic licences).

Author contribution.

Both authors contributed to the study conception, design, data analysis, and manuscript preparation. Both authors read and approved the final manuscript.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (**supplementary.pdf**) and in the anonymous artifact. Each appendix is annotated with the question it answers.

- Sup. A: Missing Test ID Sensitivity — does the absent test item change the headline gap? (**sec:appendix_missing_id**)
- Sup. B: Per-seed Extension Cells 707–1405 — does the prediction interval hold beyond the six primary seeds?
- Sup. C: Analysis Provenance — evidentiary role and multiplicity handling of each analysis. (**sec:provenance**)

- Sup. D: Canonical Checkpoint Registry — what was trained, and which families enter which claim? (`tab:checkpoint_registry`)
- Sup. E: Donor-Pool Substitution Decomposition — how much of the seen-pool advantage is pool-origin vs. similarity? (`sec:appendix_decomposition`, `tab:path_sensitivity`)
- Sup. F: Config-Faithful and Perturbation Details — protocol-mismatch check, weight-noise sweep, epoch-checkpoint decoupling search. (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`)
- Sup. G: Cluster-Aware Bootstrap — does the gap survive resampling by signer, broadcast-show, and date? (`sec:appendix_cluster_bootstrap`)
- Sup. H: Supplementary Oracle Diagnostics — is the evaluator-aware advantage robust to reference-overlap exclusion? (`sec:appendix_oracle`)
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion — does the reversal persist on a clean template-holdout split and under donor-text exclusion? (`sec:appendix_template_boundary`)
- Sup. J: Transcript-Slot Diagnostic Details — do slot-level scores reproduce the BLEU reversal? (`sec:appendix_slot`)
- Sup. K: Competence Extrapolation and Per-Item Gap Decomposition — could competence in the unobserved interval explain the reversal, and which item features drive the gaps? (`sec:appendix_extrapolation`)
- Sup. L: CSL-Daily Boundary Control — does the readout pattern generalize beyond PHX-public? (`sec:csl-daily`)
- Sup. M: Leakage Sanity Checks — can the train-pool readout be explained by implementation shortcuts? (`sec:appendix_leakage`)
- Sup. N: Pre-Exclusion Materialization Sensitivity — how sensitive are the headline numbers to the donor-exclusion rule? (`sec:appendix_materialization`)
- Sup. O: BLEURT-Substitute Validation — does a learned metric corroborate the BLEU finding under reference replacement? (`sec:appendix_bleurt`)
- Sup. P: Main-Text Tables (SI Table S22–S33) — decoded-metric sensitivity, floor effect, cross-metric gaps, leaderboard, distillation ladder, checkpoint registry, protocol summary, rescue sweep, exposure estimands, reference sensitivity, and cross-evaluator re-decoding of noisy-gloss and USTC-MoE outputs. (`sec:appendix_moved_tables`)
- Sup. Q: Additional Protocol Details — tie-breaking, scorer equivalence, metric implementations, audit-design, reference-frame, and donor-origin figures (SI Fig. S4–S6). (`sec:appendix_moved_protocol`)
- Sup. R: Reconstruction Sanity Checks (SI Table S35) — small-sample overfit test, unified inference regression, and per-checkpoint teacher-forced vs. free-decode diagnostics ruling out a systematic code-path defect. (`sec:appendix_sanity`)

Full checkpoint registry in SI Sup. D.

References

- Alkain, B., A. Núñez-Marcos, C. Escolano, L. Docío-Fernández, O. Perez-de Viñaspre, and G. Labaka. 2026. Critical analysis of datasets for sign language translation. *Frontiers in Artificial Intelligence* 9: 1743223. <https://doi.org/10.3389/frai.2026.1743223>.
- Artiaga, K., S. Kamila, H. Afli, C. Lynch, and M. Hasanuzzaman 2025. Rethinking sign language translation: The impact of signer dependence on model evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China, pp. 18379–18391. Association for Computational Linguistics.
- Camgoz, N.C., S. Hadfield, O. Koller, H. Ney, and R. Bowden 2018. Neural sign language translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7784–7793.
- Carlini, N., F. Tramèr, E. Wallace, M. Jagielski, A. Herbert-Voss, K. Lee, A. Roberts, T. Brown, D. Song, U. Erlingsson, A. Oprea, and C. Raffel 2021. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650.

- Chen, Y., J. Belouadi, and S. Eger 2022. Reproducibility issues for BERT-based evaluation metrics. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2965–2989. Association for Computational Linguistics.
- Cory, O., M. Ivashechkin, K. Sahin, O. Ranum, J. Low, E. Fish, A. Pelykh, O. Mercanoglu Sincan, and R. Bowden. 2026. BackTranslation2.0 – a linguistically motivated metric to assess sign language production. *arXiv preprint arXiv:2606.28673*. arXiv:2606.28673 .
- Czehmann, V., S. Yazdani, Y. Hamidullah, F. Nunnari, and E. Avramidis 2026. “A Sacred Bird Called the Phoenix”: Auditing the most-used parallel corpus for German sign language recognition and translation. In *Proceedings of the LREC2026 12th Workshop on the Representation and Processing of Sign Languages: Language in Motion*, Palma, Mallorca, Spain, pp. 80–92. ELRA Language Resources Association. ISBN 978-2-493814-82-1. Back-translated test set and annotations: <https://github.com/DFKI-SignLanguage/sacre-bird-phoenix> (CC BY-NC-SA 4.0).
- He, T., K. Rahul, D. Hupkes, and R. Cotterell 2024. On the blind spots of model-based evaluation metrics for text generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*.
- Hong, R. and Košecká, Jana. 2026. Conditional collapse in sign language production: A diagnostic and a scaling argument. *arXiv preprint arXiv:2606.01643*. arXiv:2606.01643 [cs.CV].
- Imai, S., M. Inan, A.B. Sicilia, and M. Alikhani 2025. SiLVERScore: Semantically-aware embeddings for sign language generation evaluation. In *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, pp. 452–461.
- Jiang, Z., C. Leong, A. Moryossef, O. Cory, M. Ivashechkin, N. Tarigopula, B. Zhang, A. Göhring, A. Rios, R. Sennrich, and S. Ebling 2025. Meaningful pose-based sign language evaluation. In *Proceedings of the Tenth Conference on Machine Translation (WMT)*, pp. 64–80.
- Kandpal, N., E. Wallace, and C. Raffel 2022. Looking inside my memory: Quantifying memorization in language models. In *Findings of EMNLP*.
- Khandelwal, U., A. Fan, D. Jurafsky, L. Zettlemoyer, and M. Lewis 2020. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*.
- Kim, J.H., M. Huerta-Enochian, C. Ko, and D.H. Lee 2024, May. SignBLEU: Automatic evaluation of multi-channel sign language translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italia, pp. 14796–14811.

ELRA and ICCL.

- Post, M. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pp. 186–191.
- Saunders, B., N.C. Camgoz, and R. Bowden 2020. Progressive transformers for end-to-end sign language production. In *European Conference on Computer Vision*.
- Saunders, B., N.C. Camgöz, and R. Bowden. 2021. Continuous 3D multi-channel sign language production via progressive transformers and mixture density networks. *International Journal of Computer Vision* 129: 2113–2135. <https://doi.org/10.1007/s11263-021-01457-9>.
- Tasyurek, S.M., T. Kiziltepe, and H.Y. Keles. 2025. Disentangle and regularize: Sign language production with articulator-based disentanglement and channel-aware regularization. *arXiv preprint arXiv:2504.06610*. arXiv:2504.06610.
- Walsh, H., E. Fish, O. Mercanoglu Sincan, M.I. Lakhal, R. Bowden, N. Fox, B. Woll, K. Wu, Z. Li, W. Zhao, H. Wang, W. Zhou, H. Li, S. Tang, J. He, X. Wang, R. Zhang, Y. Wang, L. Cheng, M. Tasyurek, T. Kiziltepe, and H.Y. Keles 2025. SLRTP2025 sign language production challenge: Methodology, results, and future work. In *Proceedings of the Third Sign Language Recognition, Translation and Production Workshop at CVPR 2025*. arXiv:2508.06951.
- Walsh, H., A. Ravanshad, M. Rahmani, and R. Bowden 2024. A data-driven representation for sign language production. In *British Machine Vision Conference*.
- Walsh, H., B. Saunders, and R. Bowden 2024. Sign stitching: A novel approach to sign language production. In *British Machine Vision Conference*.
- Yazdani, S., Y. Hamidullah, C. España-Bonet, E. Avramidis, and J. van Genabith 2026. A critical study of automatic evaluation in sign language translation. In *Proceedings of the 15th International Conference on Language Resources and Evaluation (LREC 2026)*, pp. 9535–9548.
- Zhang, T., V. Kishore, F. Wu, K.Q. Weinberger, and Y. Artzi 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*.
- Zuo, R., R.A. Potamias, E. Ververas, J. Deng, and S. Zafeiriou 2025. Signs as tokens: A retrieval-enhanced multilingual sign language generator. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.